Experiment # 6

Design Project

(This experiment should be for two to three weeks)

Goals:

To design a transistor amplifier that meets given specs.

Reference:

Microelectronics Circuit Analysis and Design, D. A. Neamen, McGraw-Hill, 4th Edition

Equipment:

Oscilloscope: Tektronix DPO 4034 Digital Oscilloscope
Function Generator: Tektronix AFG3022 Dual Channel Function Generator
Power Supply
2N2222 bipolar tranisstors
Breadboard
Capacitors available in the laboratory
Resistors available in the laboratory

Problem Statement: Design a small-signal voltage amplifier that meets the following specifications:

1) Voltage gain of 50 or more

2) Minimum input resistance of 15 kΩ

3) Output resistance less than 100 Ω

4) Maximum unclipped output of 2V peak-to-peak

5) Power supply voltage of 9 V and lower current drain from supply voltage

6) A two-stage small-signal amplifier is more cost effective than a three-stage amplifier design

Hint:

You may use a multi-stage small-signal amplifier to achieve desirable input resistance, voltage gain, output resisntace, etc.

A common-emitter amplifier with an emitter resistance R_E, but without an emitter by-pass capacitor C_E is good for high input resistance, common-emitter amplifier using an emitter by-pass capacitor is good for high voltage gain, and the use of an emitter follower results in a low output resistance.

You may use some analytical equations in Experiment 4 and Experiment 5 to help you for calculation and/or anlysis.

Design Deliverables: A design document that includes at least the following:

1) Complete electrical schematics

2) Design steps with expectations

3) Computer simulation of the design

Note:

1) Each student must design his or her own amplifier and turn the design document into the lab instructor

2) During the experiment, students can choose one of the amplifiers desinged by the members for building and testing, or can test all designs.

Experiment:

Build one of the designed amplifiers, and measure all the parameters that have been specified.

Report:

Compare desired and obtained specifications and make comments.

Your indivual report will also include the following information:
1. Laboratory partner.
2. Date and time data were taken.
3. The pre-laboratory results.

4. The experimental procedures.
5. All calculations and simulation results for each step.
6. All plots and waveforms for each step.
7. Short summary discussing what was observed for each of the steps given in experiment.
8. What you learned.